

The Total Portfolio Approach Tested Against Market Regimes:

On the Regime-Dependence of the Opportunity Cost

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Abstract

The Total Portfolio Approach (TPA) evaluates every investment through a common *opportunity cost* measured at the level of the whole portfolio. Its practical implementation, however, implicitly assumes that this opportunity cost is stable over time. We test that assumption on more than twenty years of monthly data (February 2004–May 2026) for five investable multi-asset risk factors (equity, duration, credit, carry and liquidity). A Hidden Markov Model identifies four macro-financial regimes—Goldilocks, Overheating, Recession and Stagflation—and we study how the regime-conditional opportunity cost, proxied by the conditional Sharpe ratio, reorders the factors. The ranking is far from invariant: the equity factor ranks first in Goldilocks but falls to fourth in recessions, while duration moves in the opposite direction, and the equity–bond correlation rises from 0.38 in benign states to 0.70 in stagflation, eroding diversification exactly when it is most needed. In a strict walk-forward out-of-sample experiment (no look-ahead) a regime-conditional TPA lifts the Sharpe ratio from 0.25 to 0.66 and lowers the maximum drawdown relative to its static counterpart, a difference that is statistically significant (Jobson–Korkie, $p = 0.04$). The results do not overturn the TPA; they argue for a regime-aware implementation in which relative attractiveness is assessed conditionally on the prevailing environment.

Keywords: Total Portfolio Approach; opportunity cost; market regimes; Hidden Markov Model; asset allocation; conditional diversification.

JEL: G11, G17, C58.

1 Introduction

Traditional strategic asset allocation (SAA) partitions a portfolio into asset classes managed in separate silos, each with its own return and risk targets. The Total Portfolio Approach (TPA), adopted over the last decade by several large institutional investors, replaces this logic with a single lens: every candidate investment is assessed by its marginal contribution to the total portfolio, and competing opportunities are ranked by a common *opportunity cost* (AQR Capital Management, 2023; Amundi Asset Management, 2020; Franklin Templeton, 2024). An asset is retained when the return it offers is high relative to the additional risk it brings to the portfolio as a whole.

The appeal of the framework is intuitive, but its practical deployment rests on a rarely examined premise. Because the opportunity cost is estimated from long-horizon moments—expected returns, volatilities and correlations—most implementations treat it as approximately *stationary*: a single ranking of opportunities is expected to guide allocation across the cycle. Yet

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a large body of evidence documents that the very inputs of that ranking are unstable, shifting sharply between market regimes (Ang and Bekaert, 2004; Guidolin and Timmermann, 2007). This raises a simple but consequential question, which is the object of this paper:

Is the ranking of assets by opportunity cost that underpins the TPA stable over time, or does it reverse across market regimes; and, if it does, can a regime-conditional TPA improve the consistency of allocation decisions relative to a static TPA?

We answer this question empirically. Using monthly data from February 2004 to May 2026, we construct five investable risk factors spanning the main sources of multi-asset risk and fit a four-state Hidden Markov Model (HMM) that labels each month as Goldilocks, Overheating, Recession or Stagflation. Within each regime we measure the conditional opportunity cost of every factor—proxied by its conditional Sharpe ratio—order the factors accordingly, and quantify how the ordering and the correlation structure change from one regime to the next.

The paper makes three contributions. First, it *connects two literatures that rarely intersect*: the (mostly practitioner) literature on the TPA, which assumes a stable opportunity cost, and the academic literature on market regimes, which establishes that the underlying moments are not stable. Second, it provides a *direct measurement* of the instability of the opportunity-cost ranking: the equity and duration factors swap the top and bottom of the ranking between benign and defensive regimes, and rank correlations between regimes fall as low as -0.60 . Third, it shows in a *causal, look-ahead-free walk-forward* experiment that exploiting the estimated regime—and, more pointedly, the conditional opportunity cost itself—delivers an economically and statistically significant out-of-sample improvement over static benchmarks. Throughout, the goal is not to overturn the TPA but to make its implementation regime-aware.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 formalises the opportunity cost and its regime dependence and states the hypotheses. Section 4 describes the data, factors and the HMM. Section 5 presents the empirical results, Section 6 the robustness checks, and Section 7 discusses the implementation of a conditional TPA. Section 8 concludes.

2 Related Literature

The Total Portfolio Approach. The TPA emerged among sovereign wealth and pension funds as a response to a classic limitation of siloed allocation: risks overlap across buckets, so an asset should be judged by its effect on the *whole* portfolio rather than against its own class (AQR Capital Management, 2023; Amundi Asset Management, 2020). The literature is abundant but largely produced by practitioners—asset managers, consultants and index providers—who emphasise the “what” and the “why” of the TPA (governance, factor thinking, cross-portfolio risk measurement) more than the quantitative “how” (Franklin Templeton, 2024; EDHEC-Risk Institute, 2023). Recent work has begun to formalise the approach quantitatively (Bruder et al., 2026), yet the question of the framework’s sensitivity to the market environment remains largely open. A recurring critique is that the TPA repackages ideas already present in modern portfolio theory, factor investing and risk budgeting (Markowitz, 1952; Fama and French, 1993; Maillard et al., 2010); whether it changes decisions relative to a well-built SAA is, ultimately, an empirical matter.

Market regimes. A separate strand shows that financial markets alternate between persistent states with markedly different means, volatilities and, crucially, correlations. Hamilton (1989) introduced regime switching to macro-financial time series; Ang and Bekaert (2002, 2004) and Guidolin and Timmermann (2007) demonstrate that regimes materially affect optimal asset allocation. Hidden Markov and regime-switching models are natural tools here because they

encode the *persistence* of regimes through a transition matrix (Baum et al., 1970; Rabiner, 1989), and recent applied work revisits regime-based SAA directly (Bouyé and Teïletche, 2025). The stylised fact that correlations rise in stress (Cont, 2001) is central to our diversification results.

The gap. The two literatures rarely meet. The TPA literature assumes, without testing it, that the opportunity cost is stable enough to anchor a long-horizon allocation; the regimes literature establishes that the moments on which that opportunity cost depends are not stable. To our knowledge, no study has directly measured the extent to which the *ranking of assets by opportunity cost* is reshuffled across regimes, nor drawn the implications for TPA implementation. This paper fills that gap.

3 The Opportunity Cost and Its Regime Dependence

3.1 Formalisation

Let $w \in \mathbb{R}^n$ be the vector of portfolio weights over n risk factors and Σ their covariance matrix. Total portfolio risk is $\sigma_p(w) = \sqrt{w^\top \Sigma w}$, and the marginal contribution to risk of factor i is

$$\text{MCR}_i = \frac{\partial \sigma_p}{\partial w_i} = \frac{(\Sigma w)_i}{\sqrt{w^\top \Sigma w}}. \quad (1)$$

In the spirit of the TPA, a factor is attractive to the extent that it earns a high excess return per unit of *additional* risk it imposes on the total portfolio. This motivates an opportunity-cost indicator

$$\text{OC}_i = \frac{\mu_i - r_f}{\text{MCR}_i}, \quad (2)$$

where μ_i is the expected return of factor i and r_f the risk-free rate. The higher OC_i , the more the factor deserves a place in the portfolio; ranking factors by OC_i is the operational core of a TPA decision. On short samples, marginal contributions are noisy, so we use as a robust proxy the conditional Sharpe ratio

$$\text{OC}_i \approx \text{SR}_i = \frac{\mu_i - r_f}{\sigma_i}, \quad (3)$$

and complement it with an explicit analysis of correlations to capture the diversification dimension that Eq. (3) omits.

3.2 The implicit stationarity assumption

The opportunity cost depends on three ingredients: expected returns, volatilities and correlations. The static TPA treats them as approximately constant, so that estimates from a long history can anchor the allocation. If, however, markets evolve through K latent regimes indexed by $s \in \{1, \dots, K\}$, then conditional on regime s returns have mean $\mu_i(s)$, volatility $\sigma_i(s)$ and covariance $\Sigma(s)$, and the conditional opportunity cost is

$$\text{OC}_i(s) = \frac{\mu_i(s) - r_f}{\sigma_i(s)}. \quad (4)$$

The unconditional opportunity cost used by the static TPA is a regime-probability-weighted average of the $\text{OC}_i(s)$ —and such an average can mask opposite orderings. If equity dominates in a benign regime while duration dominates in a recession, the averaged opportunity cost may suggest a spurious equivalence that no single regime validates. We therefore test two hypotheses.

H1 (ranking reversal). The ranking of factors by conditional opportunity cost $OC_i(s)$ is not invariant in s : there exist at least two regimes for which the ordering is significantly different, or even reversed.

H2 (conditional diversification). The correlation structure $\Sigma(s)$ tightens in stress regimes, reducing the diversification benefit on which the total-portfolio view rests—precisely when it is most needed.

3.3 From hypotheses to a conditional TPA

If H1 and H2 hold, using a single opportunity cost across all periods can misrank assets. A natural remedy is to condition on the estimated regime: at each date t , given the filtered regime probabilities, estimate the relevant moments and solve

$$w_t^* = \arg \max_{w \in \mathcal{W}} \left\{ w^\top \mu(\hat{s}_t) - \frac{\gamma}{2} w^\top \Sigma(\hat{s}_t) w \right\}, \quad (5)$$

with γ the risk-aversion coefficient and $\mathcal{W}$ the feasible set. We call this the *conditional TPA*. It preserves the total-portfolio logic while adapting to the observed environment. The approach is not free of costs: the current regime is never known with certainty, detection may lag turning points, and stress regimes are rare, making their conditional statistics fragile. Its value therefore rests on a trade-off between a sharper reading of the environment and the estimation noise introduced by regime detection—an essentially empirical question.

4 Data, Factors and Regime Detection

4.1 Data and factors

The analysis uses monthly total-return data from February 2004 to May 2026 ($T = 268$ observations), a window spanning the 2008 financial crisis, the euro sovereign-debt crisis, the COVID-19 shock and the 2022 return of inflation. Dollar-denominated indices are converted to euros. Five investable factors capture the main sources of multi-asset risk (Table 1). The credit factor is orthogonalised to rates by regressing the euro investment-grade index on the duration factor (beta ≈ 0.86) and keeping the residual, so that it isolates the credit premium net of the rate effect. The liquidity factor is the emerging-minus-developed equity spread, an admittedly imperfect proxy that also embeds country, currency and growth exposure; we keep it for its simplicity and investable character.

Table 1: Definition of the five investable multi-asset factors.

Factor	Underlying	Risk source
Equity	MSCI World (total return, EUR)	Equity market
Duration	Bloomberg Euro Treasury	Interest rate
Credit	Euro IG corporate, residual of duration	Credit spread
Carry	Bloomberg Euro High Yield	Carry / risk premium
Liquidity	MSCI EM – MSCI World	Liquidity / global

Over the full sample the factors differ markedly (Table 2). Equity earns the highest absolute return (9.9% p.a. for 13.1% volatility, Sharpe 0.76); duration and carry achieve comparable Sharpe ratios at much lower risk. Carry displays high kurtosis, consistent with occasional sharp corrections. Credit, net of duration, earns almost nothing over the period (Sharpe -0.06), its spread carry wiped out by 2008 and 2020, and the liquidity factor is negative (-2.1% p.a.) as emerging markets lagged since the early 2010s. These unconditional numbers are what a static investor would see—and they hide the regime dynamics that follow.

Table 2: Unconditional factor statistics, 2004–2026 (annualised).

Factor	Return	Volatility	Sharpe	Skewness	Kurtosis
Equity	9.9%	13.1%	0.76	−0.46	1.30
Duration	3.6%	4.8%	0.76	−0.91	5.61
Credit	−0.1%	1.1%	−0.06	1.33	23.93
Carry	6.9%	9.6%	0.72	−0.63	10.53
Liquidity	−2.1%	12.0%	−0.17	−0.15	−0.05

4.2 The Hidden Markov Model

To identify regimes we fit a Gaussian HMM in which observed returns depend on an unobserved state s_t , each state carrying its own mean vector and covariance. Parameters are estimated by Baum–Welch (EM) and the most likely state sequence is recovered by Viterbi (Baum et al., 1970; Rabiner, 1989). The HMM is preferred to memoryless clustering precisely because it encodes regime *persistence* through a transition matrix: over our sample the HMM produces only 14 regime switches (≈ 18 months on average), against 26 for k -means and 67 for a sign-of-growth-and-inflation rule (Table 3). It also delivers, at each date, a probability of being in each regime—directly reusable to weight allocation inputs.

Table 3: Comparison of regime-detection methods, 2004–2026.

Method	Switches	Avg. duration (m)	ARI vs. HMM	Equity Sharpe range
HMM	14	17.9	1.00	1.73
Gaussian mixture	11	22.3	0.32	2.52
k -means	26	9.9	0.38	0.90
Agglomerative	10	24.4	0.35	2.11
Quadrant rule	36	7.2	0.25	1.60

We retain $K = 4$ regimes, interpreted from the joint dynamics of growth and inflation as *Goldilocks* (solid growth, contained inflation), *Overheating* (strong growth, rising inflation), *Recession* (contracting activity, flight to safety) and *Stagflation* (weak growth, high inflation). The choice is validated formally: re-estimating the HMM for $K \in \{2, \dots, 6\}$, the Bayesian information criterion (BIC) falls steeply from $K = 2$ to $K = 4$ (from 1226 to 805) and then stops improving (it rises slightly at $K = 5$), while additional states become sparsely populated and hard to interpret (Figure 1). Goldilocks (45.1%) and Overheating (30.6%) dominate the sample; Recession (12.3%) and Stagflation (11.9%) are rarer but concentrate the main stress episodes. The estimated monthly transition matrix is strongly diagonal (self-transition probabilities above 90%), confirming the persistence that clustering cannot capture.

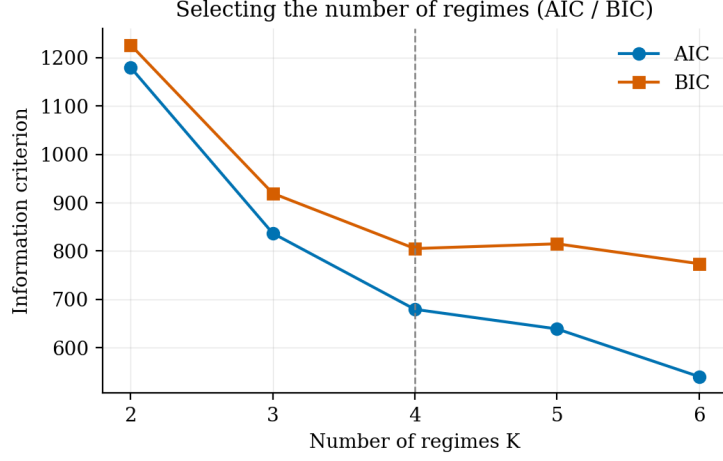


Figure 1: AIC and BIC of the HMM as a function of the number of regimes K . Most of the fit is captured by $K = 4$ (dashed line); the BIC deteriorates beyond it.

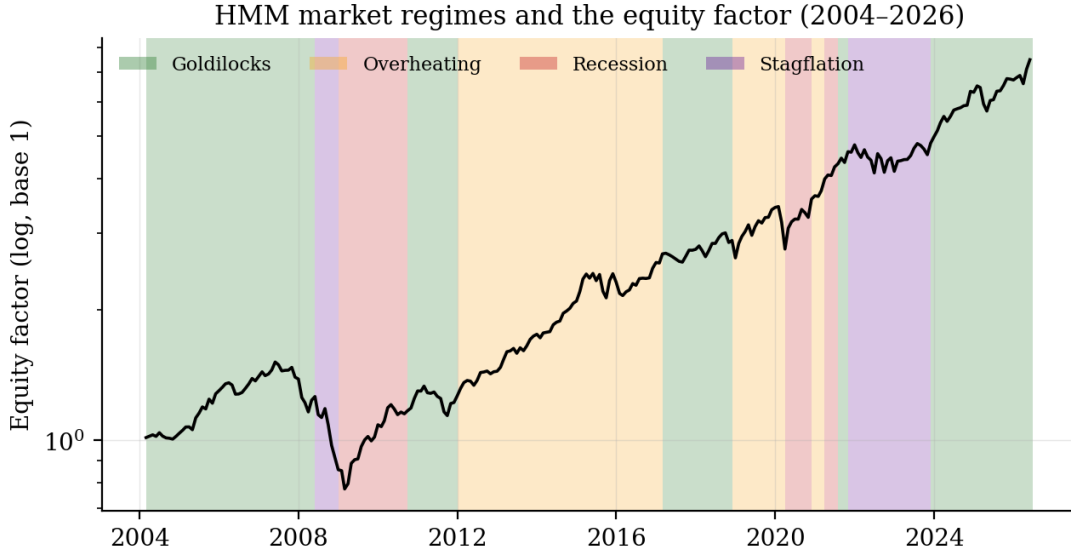


Figure 2: Estimated HMM regime sequence and the cumulative equity factor, 2004–2026. Shaded bands denote the dominant regime; stress regimes (recession, stagflation) coincide with the main drawdowns.

4.3 Protocol

To test H1 and H2 we proceed in three steps. (i) For each regime we compute the conditional Sharpe ratio of each factor, our estimator of the conditional opportunity cost. (ii) We rank the factors within each regime and measure the instability of the ordering across regimes using Spearman’s rank correlation, which is robust to the level noise in short subsamples. (iii) We compare the conditional correlation matrices, tracking in particular the equity–duration correlation as a synthetic gauge of the diversification value of a balanced portfolio. The protocol deliberately favours simple, reproducible measures over sophisticated estimators that are fragile on short samples: the aim is to demonstrate a phenomenon transparently, not to optimise a strategy.

5 Empirical Results

5.1 The conditional opportunity cost

Table 4 and Figure 3 show that factor attractiveness varies substantially across regimes. Viewed only through the unconditional column, factor differences look modest; conditionally, they are large. In Goldilocks every factor has a positive Sharpe ratio, with equity ahead (0.90). In stagflation *no* factor delivers a positive risk-adjusted return, carry being hardest hit (−1.11). Overheating and recession reshuffle the hierarchy: carry, duration and credit lead in overheating, while in recession duration (1.69) and carry (1.67) dominate and equity (0.87) loses much of its edge.

Table 4: Conditional Sharpe ratios (opportunity cost) by factor and regime.

Regime	Equity	Duration	Credit	Carry	Liquidity
Goldilocks	0.90	0.75	0.48	0.78	0.24
Overheating	1.28	1.73	−0.16	1.81	−0.76
Recession	0.87	1.69	−1.20	1.67	1.04
Stagflation	−0.47	−0.95	−0.09	−1.11	−1.46
Unconditional	0.76	0.76	−0.06	0.72	−0.17

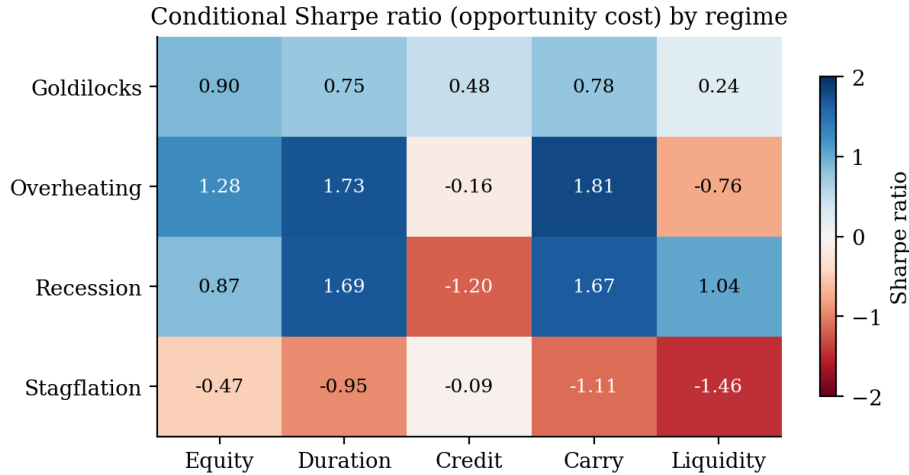


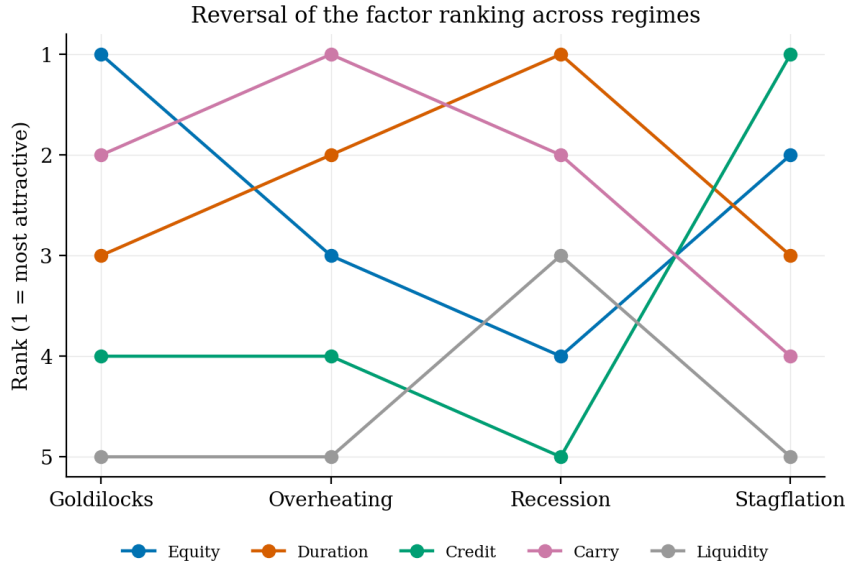
Figure 3: Conditional Sharpe ratio (opportunity cost) by factor and regime. Blue denotes attractive, red unattractive factors.

5.2 Ranking reversal: testing H1

Ordering factors by opportunity cost within each regime (Table 5, Figure 4) reveals that the ranking is *not* stable. The equity–duration pair is the clearest illustration: equity ranks first in Goldilocks and fourth in recession, while duration climbs from third to first. A single factor thus travels from the top to the bottom of the ranking as the environment changes. These reversals concentrate at major transitions—from a benign regime to a slowdown or to stagflation—and support H1: a ranking built on long-run averages may faithfully describe *none* of the regimes actually experienced.

Table 5: Ranking of factors by conditional opportunity cost (1 = most attractive).

Regime	Equity	Duration	Credit	Carry	Liquidity
Goldilocks	1	3	4	2	5
Overheating	3	2	4	1	5
Recession	4	1	5	2	3
Stagflation	2	3	1	4	5

**Figure 4:** Reversal of the factor ranking across regimes. Equity and duration swap the top and bottom of the ordering between benign and defensive regimes.

5.3 Conditional diversification: testing H2

Diversification is itself regime-dependent (Table 6, Figure 5a). The equity–duration correlation is +0.38 in Goldilocks—bonds partly absorb equity shocks—but reaches +0.70 in stagflation, where the two move together and the bond hedge loses most of its effectiveness. Diversification therefore contracts precisely in the regimes where it is most needed, validating H2. The 2022 inflation shock is the textbook case: equities and bonds fell in concert, a configuration the stagflation regime captures but that an unconditional analysis would deem improbable (Figure 5b).

Table 6: Conditional equity–duration correlation by regime.

Regime	n	Corr(Equity, Duration)	Equity vol.
Goldilocks	121	0.38	10.7%
Overheating	82	0.46	11.9%
Recession	33	0.61	17.7%
Stagflation	32	0.70	17.5%
Unconditional	268	0.55	13.1%

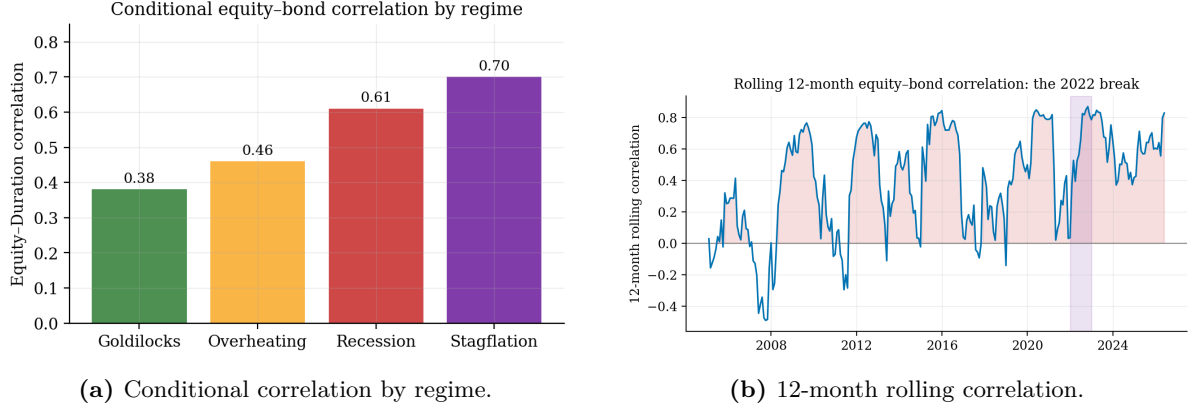


Figure 5: Equity-bond correlation. The conditional structure (left) tightens in stress; the rolling series (right) shows the 2022 structural break.

5.4 Toward a conditional TPA: a walk-forward experiment

If the opportunity-cost ranking depends on the regime, then an allocation that exploits the estimated regime should better respect TPA discipline than a static one. We compare, out-of-sample (January 2021–May 2026, 65 months), two static allocations (mean–variance, MVO; equal risk contribution, ERC) and their regime-conditional TPA counterparts, recalibrated annually. Crucially, the HMM is *re-estimated in walk-forward* at each recalibration on data available up to that point, with decisions lagged by one month, so no future information enters regime detection.

Table 7 and Figure 6 show a clear hierarchy. Conditioning lifts the Sharpe ratio from 0.25 (static MVO) to 0.66 (conditional TPA-MVO) while reducing the maximum drawdown from -10.8% to -8.1% . A properly optimised ERC reaches only 0.35: it overweights low-volatility factors (duration, credit) and is punished by the 2022 rate rise (drawdown -15.1%). The Jobson–Korkie test with the Memmel correction indicates that the gap between the conditional TPA-MVO and static MVO is statistically significant ($z = +2.1$, $p = 0.04$) (Jobson and Korkie, 1981; Memmel, 2003). Comparing the conditional TPA-MVO with a *static* TPA-MVO—same optimiser, moments fixed at the unconditional regime frequency—isolates the contribution of conditioning: the gap (0.66 vs. 0.25) is due to regime information exploited in real time, not to the factor structure alone. A rule that allocates *directly* in proportion to each factor’s conditional opportunity cost (discarding negative scores) achieves the highest Sharpe of the study (0.79, $p = 0.01$), confirming that value comes from the conditional opportunity cost itself rather than from an optimisation artefact.

Table 7: Out-of-sample performance of the allocation rules (Jan 2021–May 2026).

Allocation	Return	Volatility	Sharpe	Max. DD
Static MVO	0.9%	3.6%	0.25	-10.8%
Static ERC	1.7%	5.2%	0.35	-15.1%
Static TPA-MVO	0.9%	3.6%	0.25	-10.8%
Conditional TPA-MVO	2.5%	3.8%	0.66	-8.1%
Conditional TPA-OC	—	—	0.79	—

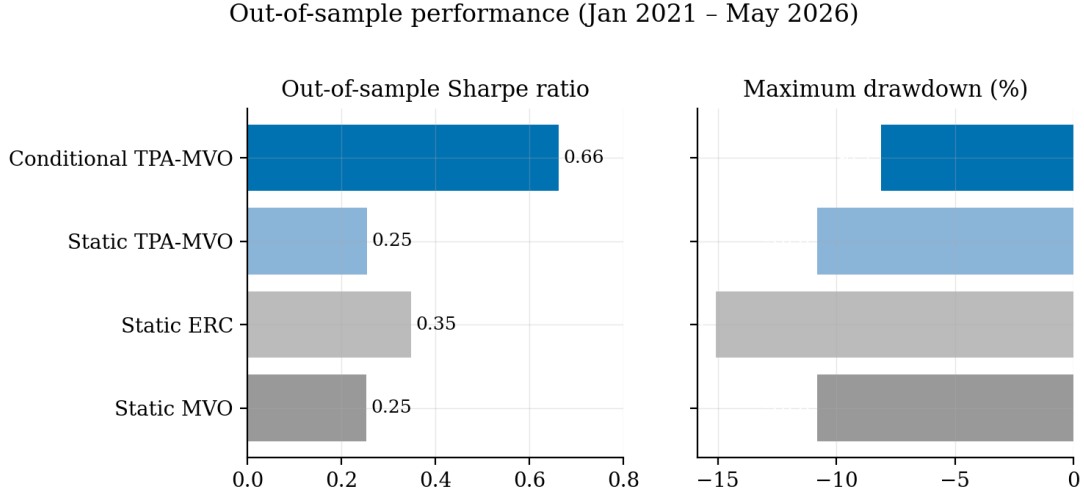


Figure 6: Out-of-sample Sharpe ratio (left) and maximum drawdown (right) of the allocation rules. The regime-conditional TPA-MVO dominates the static benchmarks.

These results must be read with caution: the out-of-sample window is short (65 months, six annual recalibrations), so the significance attests to a real gain *over the period* without prejudging future performance.

6 Robustness

Three checks support the findings. First, *rank stability*: Spearman rank correlations between the opportunity-cost rankings of different regimes range from -0.60 to $+0.70$, the most negative being stagflation vs. recession (Table 8); the reversals are concentrated, not diffuse. Second, *estimation uncertainty*: bootstrapping the conditional Sharpe ratios (5,000 resamples within each regime) yields wide intervals for the rare stress regimes but confirms the equity/duration reversal—in recession, the duration Sharpe stays clearly positive (90% CI $[+0.52, +3.94]$) while the equity interval straddles zero—which is why we rely on rank measures. Third, *model selection*: the BIC confirms $K = 4$ (Section 4), and alternative detectors (k -means, Gaussian mixture, agglomerative, quadrant rule) change the regime chronology but not the qualitative conclusion.

Table 8: Spearman rank correlation of the opportunity-cost rankings between regimes.

	Goldilocks	Overheating	Recession	Stagflation
Goldilocks	1.00	0.70	0.10	0.30
Overheating	0.70	1.00	0.60	0.00
Recession	0.10	0.60	1.00	-0.60
Stagflation	0.30	0.00	-0.60	1.00

7 Discussion: Implementing a Conditional TPA

The evidence does not challenge the principle of the TPA; it qualifies the stationarity premise on which its implementation usually rests. Returns, volatilities and correlations move with the environment, so a single set of inputs cannot describe every phase. The TPA is best seen as a decision framework rather than a fixed allocation—one that gains from explicitly incorporating the regime in which the investor operates.

Operationally, conditioning need not mean constant rebalancing. Regime probabilities are one additional input to the decision process: tilt toward equity in benign phases, toward duration and credit in slowdowns, and reduce overall risk in the most degraded states. Two caveats temper the enthusiasm. The current regime is never known with certainty and detection can lag turning points; and stress regimes are rare, so their conditional statistics are fragile. A conditional TPA should thus support—not replace—judgement, remaining legible enough for investment committees to interpret its signals, especially when they are uncertain or conflicting.

Several extensions are natural: incorporating private assets, whose returns are harder to characterise; testing alternative regime detectors and the impact of transaction costs and reaction lags; and studying the interaction between regimes and the specific constraints of regulated long-term investors.

8 Conclusion

This paper asked whether the opportunity cost at the heart of the Total Portfolio Approach is stable across market environments. Over more than twenty years of data it is not. Risk-adjusted returns, correlations and the resulting ranking of factors shift sharply across regimes: equity and duration exchange the top and bottom of the ranking between benign and defensive states, and equity–bond diversification collapses in stagflation. In a strict, look-ahead-free walk-forward experiment, a regime-conditional TPA is statistically superior to its static counterpart over the sample ($p = 0.04$), with an even larger gain when the allocation is driven directly by the conditional opportunity cost. These findings do not overturn the TPA; they show that market conditions strongly shape the relative attractiveness of risk factors and that a regime-aware implementation better preserves the discipline the framework is meant to enforce. The short out-of-sample window counsels caution before extrapolating the performance gain, but the qualitative message—condition the opportunity cost on the regime—is robust.

References

- Amundi Asset Management. Multi-asset total portfolio approach in a complex world. Technical report, Amundi, 2020.
- Andrew Ang and Geert Bekaert. Regime switches in interest rates. *Journal of Business & Economic Statistics*, 20(2):163–182, 2002.
- Andrew Ang and Geert Bekaert. How regimes affect asset allocation. *Financial Analysts Journal*, 60(2):86–99, 2004.
- AQR Capital Management. The total portfolio approach. Technical report, AQR, Alternative Thinking, 2023.
- Leonard E. Baum, Ted Petrie, George Soules, and Norman Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171, 1970.
- Eric Bouyé and Jérôme Teïletche. *Regime-Based Strategic Asset Allocation*. Taylor & Francis, 2025.
- Benoît Bruder, Nicolas Gaussel, Jean-Charles Richard, and Thierry Roncalli. A quantitative framework for the total portfolio approach. *arXiv preprint arXiv:2603.03213*, 2026.
- Rama Cont. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1:223–236, 2001.

- EDHEC-Risk Institute. Total portfolio approach and private assets, parts 1–3. Technical report, EDHEC Business School, 2023.
- Eugene F. Fama and Kenneth R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993.
- Franklin Templeton. The rise of total portfolio approach. Technical report, Franklin Templeton Investment Management, 2024.
- Massimo Guidolin and Allan Timmermann. Asset allocation under multivariate regime switching. *Journal of Economic Dynamics and Control*, 31(11):3503–3544, 2007.
- James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.
- J. D. Jobson and Bob M. Korkie. Performance hypothesis testing with the Sharpe and Treynor measures. *The Journal of Finance*, 36(4):889–908, 1981.
- Sébastien Maillard, Thierry Roncalli, and Jérôme Teïletche. The properties of equally weighted risk contribution portfolios. *The Journal of Portfolio Management*, 36(4):60–70, 2010.
- Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.
- Christoph Memmel. Performance hypothesis testing with the Sharpe ratio. *Finance Letters*, 1: 21–23, 2003.
- Lawrence R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.